COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:		**SECTION:**	
DATE PERFORMED:		**DATE SUBMITTED:**	
NAME:		**INSTRUCTOR:**	

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILO A: Implement static and dynamic memory allocation & ILO B: Create dynamically allocated objects using pointers and arrays In this activity, we will explore static and dynamic memory allocation through by utilizing arrays and other components mentioned beforehand, this will be for the implementation of a student list with the students' names and age. To begin, we will include pertinent libraries for input/output stream and strings (which we will be using in this activity).

6. Output:

Screenshot	
Observation	

Table 2-1. Initial Driver Program

Screenshot	
Observation	

Table 2-2. Modified Driver Program with Student Lists

Loop A	
Observation	
Loop B	

Observation	
Output	
Observation	

Table 2-3. Final Driver Program

Modification	
Observation	

Table 2-4. Modifications/Corrections Necessary

7. Supplementary Activity

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ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers

Jenna's Grocery List

Apple	PHP 10	x7
Banana	PHP 10	x8
Broccoli	PHP 60	x12
Lettuce	PHP 50	x10

Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.

Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.

Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.

Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.

Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.

Problem 5: Discuss the Provide the following:

- Summary of lessons learned
- Analysis of the procedure
- Analysis of the supplementary activity
- Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

Output:

```
PS C:\Users\solvendar\OneDrive\Dokumen\CPE21S4_BAL>  & 'c:\
oft-MIEngine-In-3bigishc.bfr' '--stdout=Microsoft-MIEngine-
\ucrt64\bin\gdb.exe' '--interpreter=mi'
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.
Jenna's Grocery List

Fruit Name: Apple
Price: 10
Quantity: 7
Total :70
Apple 10 7
Fruit Name: Banana
Price: 10
Quantity: 8
Total :80
Banana 10 8
Vegetable Name: Brocolli
Price: 60
Quantity: 12
Total :720
Brocolli 60 12
Vegetable Name: Lettuce
Price: 50
Quantity: 10
Total :500
Lettuce 50 10

 Total Amount to Pay: PHP 1370
Destructor Called.
Fruit Name: Apple
Price: 10
Quantity: 7
Total :70
Apple 10 7
Fruit Name: Banana
Price: 10
Quantity: 8
Total :80
Banana 10 8
Vegetable Name: Brocolli
Price: 60
Quantity: 12
Total :720
Brocolli 60 12
Empty Slot. The Element at the index 3 was deleted.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\solvendar\OneDrive\Dokumen\CPE21S4_BAL>
```

Code Analysis: I created a 2 classes which is fruit and vegetables, and their attributes are private like their name, price, and quantity and on the public, i declared a setter and getter for the private attributes and created a constructor which is come from the setter and the deconstructor for clearing all the memory after. I created a sum method and placed to the fruits and vegetables that return the total for the vegetable and fruits that i created a method below them and for the copy assignment in this code (this==&copyFruit), the function checks first for the self assignment to it to make sure the object is not accidentally trying to copy itself and saying “copy assignment has been called” and if does and its a valid copy, it will copy each individual private attributes like the name, price, and quantity from the source object to the current object and then it returns the reference to itself (return *this) so the chained assignment can work properly. And in the total sum price, i created a function that inside was the class for the fruitlist1 and veglist2 and assigning values to it using the parameters constructor, it handles

displaying the fruit and vegetables by looping through both arrays and calling printDetails() on each element to print the attributes of them and also the subtotals. After that it will move to the destructor 1 which is the veglist2[1].~Vegetables which is the lettuce that executes the destructor and prints the corresponding. And in redisplaying items and veglist2[0] and the empty slot of it by looping through the remaining fruit items and clearly printing the broccoli via veglist2[0] to prove and see that it was still untouched, and after there's a custom message that saying that the element of the index 3 was deleted.

Conclusion: I learned in this activity that how encapsulation work when it comes this kind of complex activity for me, like separating the vegetable and the fruits and also declaring a setter and getter of each class since the attribute of each class was private and need to access it and i accessed it by public getters and also adding a method where the sum of the price of an item and also the quantity of it and using a special functions like constructor, destructor, etc. and also how to manage objects and handling memory states from creating and deleting it to avoid memory leaking. I want to learn more about this topic before hopping to linked lists, stacks, queues, etc. It enhances my logical thinking when handling this kind of activity.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini AI Search

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code Assistance, Explaining the concepts, and idea generation

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: Asking about the complex part that I couldn't understand when i search it in the explaining the concepts part and also how does the flow run when it comes to the complex part like adding the copy assignment operator and give a little lines of code that separate to the hands-on and example line of that part only so I can have an idea how to use it properly and for the code assistance is being my code checker of why does it compile error and say something that I'll figure it out by myself (ex. your constructor, you are trying to stuff Jenna's entire shopping list into the default arguments using syntax like {(7, 8)} and std::string{("Apple"),("Banana")}. This will cause a compiler error.)

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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